

Sensing-aided Multi-modal Channel Prediction in 6G mmWave Massive MIMO Systems

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Abstract—To integrate a variety of data-demanding and delay-sensitive applications, millimeter-wave (mmWave) massive multiple-input multiple-output (MIMO) system has emerged as a key enabler for 6G wireless communications. To fully exploit mmWave massive MIMO system, acquisition of accurate channel information is of great importance while such is challenging due to short coherence time of mmWave channel. Therefore, to obtain accurate channel information in a real-time manner, we propose sensing-aided multi-modal channel prediction technique (SMCPT). By analyzing sensing images, we can accurately and quickly extract the positions of mobile devices as well as the positions, orientations, and materials of entities affecting signal propagation, thus enhancing channel prediction accuracy. From practical experiments, we show that SMCPT achieves more than 69% channel prediction accuracy gain over conventional schemes.

I. INTRODUCTION

To support a variety of data-demanding and delay-sensitive applications in 6G, millimeter-wave (mmWave) massive multiple-input multiple-output (MIMO) system has received much attention recently [1]. Recently, world radio conference (WRC) identified three potential spectrum bands for 6G (4.4~4.8 GHz, 7.1~8.5 GHz, and 14.8~15.3 GHz). By exploiting abundant spectrum resources and a large number of antennas, mmWave massive MIMO system can realize future-oriented 6G services such as AR/XR, autonomous driving, and high-fidelity holographic displays [2]. Essence of beamforming is to differently shift the signal phases across antennas to overcome the severe signal attenuation of mmWave signals, so its efficacy relies entirely on the precise arrangement of signals across antennas. Without doubt, precise and timely acquisition of channel information is of great importance.

Traditionally, linear channel estimation techniques have been widely used to acquire channel information [3]. However, in mmWave systems where the carrier frequencies are significantly high, channel coherence time is very short, causing a channel variation even in the middle of the channel estimation period. To mitigate the performance loss caused by the so-called channel aging effect, various channel prediction techniques have been proposed [4], [5]. In [4], the channel

prediction technique using sinusoidal channel approximation has been proposed. In [5], deep learning (DL)-based channel prediction techniques have been proposed. Since the conventional channel prediction techniques rely solely on the previously obtained channel information, it is very difficult, and in fact not possible, to successfully predict the channel when channel dynamics change over time. For example, when a user suddenly moves or the line-of-sight (LoS) link is suddenly blocked by obstacles, conventional channel prediction techniques fail to estimate the future channel since they rely on the past channels obtained under a different channel environment. Thus, predicting channels solely based on the past estimated channels is insufficient for high-frequency band signals in mmWave massive MIMO systems.

The main goal of this paper is to introduce a sensing-aided framework in mmWave channel prediction. Recently, we are witnessing an astonishing success of various sensing-based computer vision (CV) techniques in object detection, image segmentation, and crowd estimation [6]. Motivated by this, we propose a technique called *sensing-aided multi-modal channel prediction technique* (SMCPT) that combines vision sensing and conventional RF signaling to quickly and accurately predict the real-time channel information. Specifically, leveraging the powerful capability of DL techniques in solving the inverse problems (i.e., reconstructing an unknown information from observations), deep neural network (DNN) in SMCPT identifies wireless entities (e.g., mobile devices and scatterers) using the sensing images and past RF-based estimated channels. Then, by reconstructing the wireless environment with the identified entities, SMCPT can precisely identify the channel parameters (e.g., path gain, delay, and angles) for each channel propagation path and thereby accurately predict the channel. By analyzing sensing images, we can extract various geometric and contextual information including the positions, orientations, and materials of the entities (e.g., humans, vehicles, walls, and ground). Note, due to the short capture time of RGB cameras (around 1 ms including exposure time and ADC processing time), we can analyze surrounding environment and track the channel with minimal latency.

SMCPT largely consists of two DNNs. First, using *channel parameter prediction network* (CPPN), we initially predict the future geometric channel parameters. These channel parameters provide the geographical distribution of channel paths

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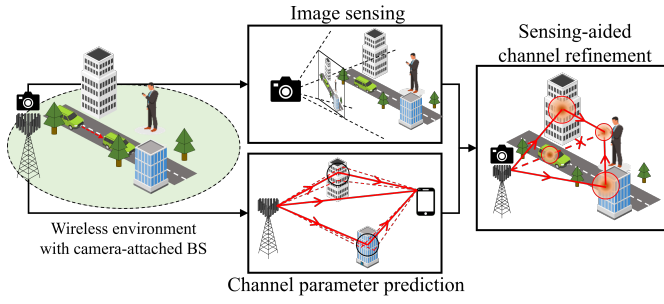


Fig. 1: Conceptual diagram of the proposed SMCPT.

between BS and mobile device. Second, using *sensing-aided channel refinement network* (SCRN), we combine the initially predicted channel parameters and sensing images to refine the channel prediction. Key structure of SCRN is cross-attention that quantifies the correlations between two different inputs and differently weighs the elements of one input based on the measured correlations. Using the cross-attention mechanism with initially predicted channel parameters and sensing images, SCRN can find out and highlight the pixel regions that contain the spatial information of propagation environment. By focusing on these pixel regions, BS can accurately extract the exact positions of mobile devices and also the positions and orientations of scatterers and possible obstacles.

II. MMWAVE MASSIVE MIMO SYSTEMS

A. System Model

We consider a downlink mmWave massive MISO communication system where BS equipped with $N_t = N_{t,x} \times N_{t,y}$ uniform planar array (UPA) antenna and an RGB camera serves a single-antenna user. Under this setup, the received signal of the user at time t is

$$y_t = \mathbf{h}_t^H \mathbf{f}_t s_t + n_t \in \mathbb{C}, \quad (1)$$

where $\mathbf{h}_t \in \mathbb{C}^N$ is the downlink mmWave channel, $\mathbf{f}_t \in \mathbb{C}^N$ is the precoding vector, and $n_t \sim \mathcal{CN}(0, \sigma_n^2)$ is Gaussian noise. Then, the achievable data rate is $R = \log_2 \left(1 + \frac{|\mathbf{h}_t^H \mathbf{f}_t|^2}{\sigma_n^2} \right)$.

We use the geometric multipath channel model as

$$\mathbf{h}_t = \sum_{\ell=1}^L \sqrt{\beta_{t,\ell}} e^{-j2\pi f_c \tau_{t,\ell}} \mathbf{a}(\phi_{t,\ell}, \theta_{t,\ell}) \in \mathbb{C}^N, \quad (2)$$

where L is number of propagation paths, $\beta_{t,\ell}$ is path gain, f_c is carrier frequency, $\tau_{t,\ell}$ is path delay, and $\phi_{t,\ell} \in [-\pi, \pi)$ and $\theta_{t,\ell} \in [0, \pi)$ are the angle of departure (AoD) of azimuth and elevation, respectively. Also, $\mathbf{a}(\phi, \theta)$ is the UPA antenna array response vector expressed as $\mathbf{a}(\phi, \theta) = \mathbf{a}_h(\phi, \theta) \otimes \mathbf{a}_v(\theta)$, where $\mathbf{a}_h(\phi, \theta) = [1, e^{-j\pi \cos \phi \sin \theta}, \dots, e^{-j\pi(N_x-1) \cos \phi \sin \theta}]^T$, $\mathbf{a}_v(\theta) = [1, e^{-j\pi \cos \theta}, \dots, e^{-j\pi(N_y-1) \cos \theta}]^T$. Note that $\mathbf{h}_t$ can be succinctly expressed as a function of channel parameters:

$$\mathbf{h}_t = \mathbf{A}(\phi_t, \theta_t) \text{diag}(\beta_t) \exp(-2\pi f_c \tau_t), \quad (3)$$

where $\phi_t = [\phi_{t,1}, \dots, \phi_{t,L}]^T$, $\theta_t = [\theta_{t,1}, \dots, \theta_{t,L}]^T$, $\beta_t = [\beta_{t,1}, \dots, \beta_{t,L}]^T$, $\tau_t = [\tau_{t,1}, \dots, \tau_{t,L}]^T$, $\mathbf{A}(\phi, \theta) =$

$[\mathbf{a}(\phi_1, \theta_1), \dots, \mathbf{a}(\phi_L, \theta_L)] \in \mathbb{C}^{N \times L}$, and $\exp(\mathbf{x}) = [e^{x_1}, \dots, e^{x_L}]^T$ is the element-wise exponential operation. Then, we define the parametric channel state information (CSI) $\mathbf{P}_t$ as $\mathbf{P}_t = [\phi_t, \theta_t, \beta_t, \tau_t]^T \in \mathbb{R}^{4 \times L}$.

B. Conventional Channel Prediction Techniques

To overcome the channel aging effect, a variety of channel prediction techniques have been proposed [4], [5]. Due to sole reliance on channel information obtained in the past, they may fail to predict unexpected changes in the channel. For example, the conventional techniques cannot take into account sudden events such as movement direction change, hand tremor, or blockage. In such cases, a severe degradation of channel prediction accuracy is inevitable with the conventional channel prediction techniques relying on past channel features.

To capture the dynamic channel variation, various camera sensing-based channel prediction techniques have been proposed recently [6]–[8]. While one can instantaneously find out the directions of mobile devices from BS with camera sensing, when the LoS path is blocked by obstacles, the acquisition of channel information through such single-modal sensing becomes challenging. Since we cannot obtain any information on the position of mobile device and NLoS path scatterers, it is impossible to extract channel-related visual information from sensing images, thereby failing to predict channel.

III. SENSING-AIDED MULTI-MODAL CHANNEL PREDICTION TECHNIQUE

An intriguing feature of the proposed SMCPT is to predict the channels using both the past channels and sensing images. In a nutshell, the proposed SMCPT uses the M past channel estimations $\hat{\mathbf{h}}_{-(M-1)}, \hat{\mathbf{h}}_{-(M-2)}, \dots, \hat{\mathbf{h}}_0$ and the sensing images $\mathcal{I}_d, \mathcal{I}_{2d}, \dots, \mathcal{I}_{Nd}$ to predict the N channels $\bar{\mathbf{h}}_d, \bar{\mathbf{h}}_{2d}, \dots, \bar{\mathbf{h}}_{Nd}$ ¹. Major benefit of SMCPT is that we can capture both large-scale and small-scale channel variations by exploiting past estimated channels and sensing images and thereby accurately track the channel in a real-time manner. Specifically, using the past RF-based estimated channels, we can extract the channel variation pattern for each mobile device in terms of channel parameters (e.g., path gain, delay, and angles) that reveal the long-term channel variation. Using this as prior information, we can determine where to focus on in the sensing images to extract the position of each mobile device and also indicate the positions and orientations of scatterers and blockages associated with each device. By immediately detecting and analyzing the signal propagation environment with camera sensing images, we can quickly identify the wireless environments including unexpected events (e.g., blockage, user movement pattern change, and hand tremor), thereby enhancing the channel prediction accuracy. The proposed SMCPT consists of two major steps: *channel parameter prediction* and *sensing-aided channel refinement* (see Fig. 2).

¹We can adjust the value of d so that the time difference between the predicted channels is less than the channel coherence time T_c , ensuring that the predicted channel is coherent for every slot within the frame.

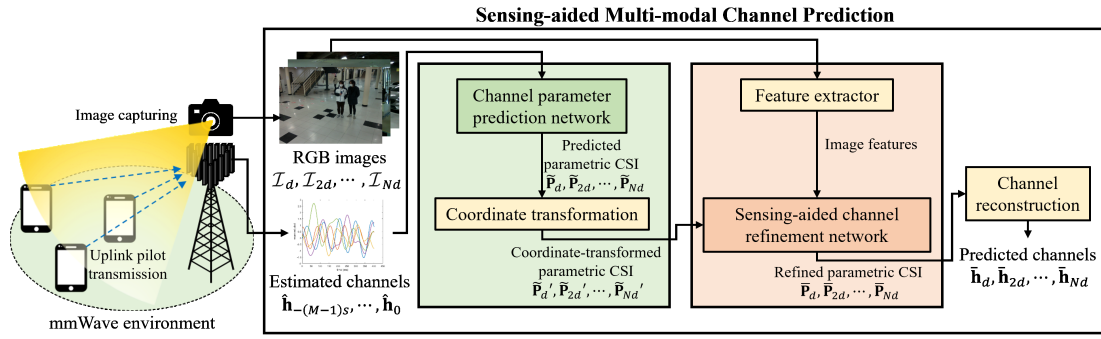


Fig. 2: Overall operations of the proposed SMCPT.

A. Channel Parameter Prediction

For the channel parameter prediction, we use CPPN, a Transformer-based neural network dedicated to extracting the temporal correlations in the historical channels to initially predict the parametric CSI $\tilde{\mathbf{P}}_d, \tilde{\mathbf{P}}_{2d}, \dots, \tilde{\mathbf{P}}_{Nd}$ where

$$\tilde{\mathbf{P}}_{nd} = [\tilde{s}_{nd}, \tilde{\phi}_{nd}, \tilde{\theta}_{nd}, \tilde{\beta}_{nd}, \tilde{\tau}_{nd}]^T \in \mathbb{R}^{5 \times L_{\max}}, \quad (4)$$

where $L_{\max}$ is the maximum number of paths, $\tilde{\phi}_{nd}, \tilde{\theta}_{nd}, \tilde{\beta}_{nd}, \tilde{\tau}_{nd}$ are the initial channel parameter predictions, and $\tilde{s}_{nd} \in [0, 1]^{L_{\max}}$ is the score vector that indicates the probability of each path being a valid one².

In CPPN, a series of temporal Transformer blocks perform attention operations to determine how much each past channel is relevant to current channels. In a temporal Transformer block, for k -th attention head, we obtain the three different embedding matrices for k -th attention head, i.e., query $\mathbf{Q}_k = \mathbf{W}_k^Q \mathbf{U} \in \mathbb{R}^{(D/N_h) \times M}$, key $\mathbf{K}_k = \mathbf{W}_k^K \mathbf{U} \in \mathbb{R}^{(D/N_h) \times M}$, and value $\mathbf{V}_k = \mathbf{W}_k^V \mathbf{U} \in \mathbb{R}^{(D/N_h) \times M}$ where $\mathbf{W}_k^Q, \mathbf{W}_k^K, \mathbf{W}_k^V \in \mathbb{R}^{(D/N_h) \times D}$ are matrices of trainable parameters and N_h is the number of attention heads. Using $\mathbf{Q}_k$ and $\mathbf{K}_k$, the attention map is calculated as

$$\mathbf{M}_k = f_{\text{softmax}} \left(\frac{\mathbf{K}_k^T \mathbf{Q}_k}{\sqrt{D/N_h}} \right) \in \mathbb{R}^{M \times M}, \quad (5)$$

where f_{softmax} is column-wise softmax function defined as $[f_{\text{softmax}}(\mathbf{X})]_{i,j} = \frac{e^{[\mathbf{X}]_{i,j}}}{\sum_i e^{[\mathbf{X}]_{i,j}}}$. The attention map $\mathbf{M}_k$ determines how much each past channel is relevant to current channel. Then, by performing the weighted sum of the value matrix $\mathbf{V}_k$ with the weights in $\mathbf{M}_k$, we obtain context matrix $\mathbf{C}_k = \mathbf{V}_k \mathbf{M}_k \in \mathbb{R}^{(D/N_h) \times M}$. Finally, by passing the output of the last temporal Transformer block through a fully connected layer, the parametric CSI prediction $\tilde{\mathbf{P}}$ is obtained.

B. Image Acquisition and Coordinate Transformation

To exploit vision sensing in channel prediction, sensing direction should be determined properly so that the captured image contains meaningful information on channel. To do so, BS can leverage the SSB beam index obtained at the initial access process. During the initial access, BS transmits a set of

SSB beams, each of which covers specific physical area. Then the mobile measures RSRP of the SSB beams and then feeds back the index of the SSB beam corresponding to the largest RSRP. Based on this observation, BS captures the image of the area covered by the SSB beam index.

Then, to utilize the images captured from the SSB beam direction, we need to transform the AoDs of the channel paths expressed with respect to the antenna orientation into the angles with respect to the camera orientation (see Fig. 3). This transformation allows us to identify and emphasize the regions within the RGB images that are relevant to the channel paths, thereby making the vision sensing an effective means assisting the channel prediction refinement.

Proposition 1. The AoDs (ϕ, θ) are transformed into the angles (ϕ', θ') in the camera coordinate system as

$$\phi' = \arccos \left(\frac{-\sin \theta \sin(\phi - \phi_0)}{\sqrt{1 - (\sin \theta \cos \theta_0 \cos(\phi - \phi_0) - \cos \theta \sin \theta_0)^2}} \right), \quad (6)$$

$$\theta' = \arccos(\sin \theta \sin \theta_0 \cos(\phi - \phi_0) + \cos \theta \cos \theta_0). \quad (7)$$

Proof. Consider the unit normal vector of camera lens plane $\mathbf{v} = [\sin \theta_0 \cos \phi_0, \sin \theta_0 \sin \phi_0, \cos \theta_0]$. Then, the horizontal and vertical directions of camera lens are

$$\mathbf{u} = \frac{\mathbf{v} \times \mathbf{z}}{\|\mathbf{v} \times \mathbf{z}\|} = [\sin \phi_0, -\cos \phi_0, 0], \quad (8)$$

$$\mathbf{w} = \frac{\mathbf{u} \times \mathbf{v}}{\|\mathbf{u} \times \mathbf{v}\|} = [-\cos \theta_0 \cos \phi_0, -\cos \theta_0 \sin \phi_0, \sin \theta_0], \quad (9)$$

respectively. Then, consider unit vector $\mathbf{p} = [\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta]$ pointing in (ϕ, θ) . By expressing $\mathbf{p}$ using the basis defined by $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ and then using the Cartesian-to-spherical coordinate transformation, we can easily obtain (6) and (7). $\square$

Applying (6) and (7) to (ϕ_ℓ, θ_ℓ) , we obtain the coordinate-transformed parametric CSI as $\tilde{\mathbf{P}}'_{nd} = [\tilde{\phi}'_{nd}, \tilde{\theta}'_{nd}, \tilde{\beta}_{nd}, \tilde{\tau}_{nd}]^T$.

C. Sensing-aided Channel Refinement

For the sensing-aided channel refinement, we use SCRNN, another Transformer-based network that combines the initial parametric CSI prediction $\tilde{\mathbf{P}}'_{nd}$ and the RGB image $\mathcal{I}_{nd}$ to obtain an enhanced parametric CSI prediction

$$\bar{\mathbf{P}}_{nd} = [\bar{s}_{nd}, \bar{\phi}_{nd}, \bar{\theta}_{nd}, \bar{\beta}_{nd}, \bar{\tau}_{nd}]^T \in \mathbb{R}^{5 \times L_{\max}}, \quad (10)$$

²To keep the dimension of CPPN constant even with varying number of channel paths, we make $L_{\max}$ parameter prediction sets and then additionally predict a score vector $\tilde{s}_{nd} = [\tilde{s}_{nd,1}, \dots, \tilde{s}_{nd,L_{\max}}]^T$.

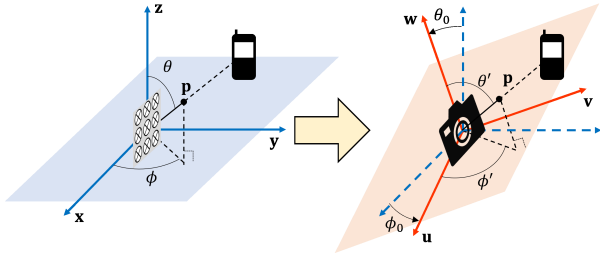


Fig. 3: Coordinate transformation.

where $\bar{\phi}_{nd}, \bar{\theta}_{nd}, \bar{\beta}_{nd}, \bar{\tau}_{nd}$ are the final channel parameter predictions and $\bar{s}_{nd} \in [0, 1]^{L_{\max}}$ is the score vector as in CPPN.

Before SCRNN, we employ a feature extractor to convert the input image $\mathcal{I} \in \mathbb{R}^{H \times W \times 3}$ into image features $\mathbf{F} \in \mathbb{R}^{D \times S}$ where S is an embedding dimension³.

In SCRNN, a series of fusion Transformer blocks perform cross-attention operations to extract the spatially correlated features between two inputs and thus find out the image regions that are highly correlated with the channel parameters. In a fusion Transformer block, for k -th attention head, we obtain three different embedding matrices, i.e., query $\mathbf{Q}_k^\dagger = \mathbf{W}_k^{Q^\dagger} \mathbf{S} \in \mathbb{R}^{(D/N_h) \times L}$, key $\mathbf{K}_k^\dagger = \mathbf{W}_k^{K^\dagger} \mathbf{F} \in \mathbb{R}^{(D/N_h) \times S}$, and value $\mathbf{V}_k^\dagger = \mathbf{W}_k^{V^\dagger} \mathbf{F} \in \mathbb{R}^{(D/N_h) \times S}$ where $\mathbf{W}_k^{Q^\dagger}, \mathbf{W}_k^{K^\dagger}, \mathbf{W}_k^{V^\dagger} \in \mathbb{R}^{(D/N_h) \times D}$ are matrices of trainable parameters. Using $\mathbf{Q}_k^\dagger$ and $\mathbf{K}_k^\dagger$, the cross-attention map is calculated as

$$\mathbf{M}_k^\dagger = f_{\text{softmax}} \left(\frac{\mathbf{K}_k^{\dagger T} \mathbf{Q}_k^\dagger}{\sqrt{D/N_h}} \right) \in \mathbb{R}^{S \times L}. \quad (11)$$

The cross-attention map $\mathbf{M}_k^\dagger$ determines the importance of each pixel in identifying the wireless entities in the environment. Then, by performing the weighted sum of the value matrix $\mathbf{V}_k^\dagger$ with weights determined by $\mathbf{M}_k^\dagger$, we obtain the context matrix $\mathbf{C}_k^\dagger = \mathbf{V}_k^\dagger \mathbf{M}_k^\dagger \in \mathbb{R}^{(D/N_h) \times L}$. Finally, by passing the output of the last fusion Transformer block through a fully-connected layer, the refined parametric CSI prediction $\bar{\mathbf{P}}'$ is obtained.

Then, by performing the reverse of coordinate transformation in **Proposition 1** to $\bar{\mathbf{P}}'$, we can obtain the refined channel parameter predictions $\bar{\mathbf{P}} = [\bar{s}, \bar{\phi}, \bar{\theta}, \bar{\beta}, \bar{\tau}]^T$. Finally, the refined channel predictions can be obtained as

$$\bar{\mathbf{h}} = \mathbf{A}(\bar{\phi}, \bar{\theta}) \text{diag}(\mathbf{1}_{\bar{s} \geq 0.5}^T \bar{\beta}) \exp(-2\pi f_c \bar{\tau}). \quad (12)$$

D. SMCPT Network Training

In training CPPN and SCRNN, we update the network parameters δ such that the network can generate the optimal channel parameter predictions $\bar{\mathbf{P}}$ using the supervised learning with the ground-truth channel parameters. The ground-truth channel parameters are obtained through ray tracing techniques in the simulation environment. To determine which ground-truth path components correspond to which predicted path components, we define the mapping function $\sigma : \{1, \dots, L\} \rightarrow \{1, \dots, L_{\max}\}$ and then find out the optimal matching σ^* that minimizes the difference between the predicted parameters and

³For simplicity, we omit the subscript denoting the time instant.

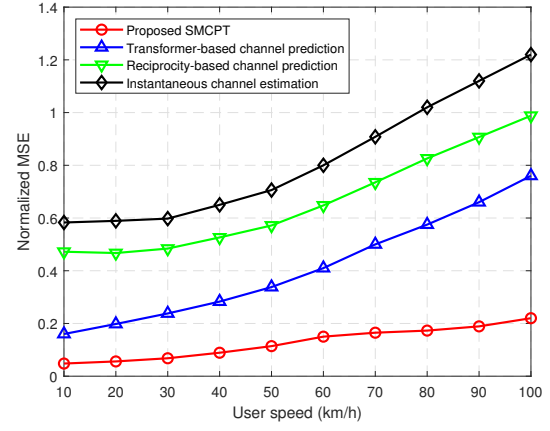


Fig. 4: Channel NMSE vs. user speed (SNR = 10).

true parameters as $\sigma^* = \text{argmin}_{\sigma} \sum_{\ell=1}^L (|\bar{\beta}_{\sigma(\ell)} - \beta_{\ell}| + |\bar{\tau}_{\sigma(\ell)} - \tau_{\ell}| + |\bar{\phi}_{\sigma(\ell)} - \phi_{\ell}| + |\bar{\theta}_{\sigma(\ell)} - \theta_{\ell}|)$. The optimal matching σ^* can be found using the Hungarian matching algorithm [9].

After obtaining the optimal matching function σ^* , we train the network parameters δ to minimize the regression loss function $\mathcal{L}_{\delta}$ that quantifies the difference between the predicted channel parameters $\bar{\mathbf{P}}_{\sigma^*}$ reordered by the matching function σ^* and the true channel parameters $\mathbf{P}$. Specifically, the loss function $\mathcal{L}_{\delta}(\bar{\mathbf{P}}_{\sigma^*}, \mathbf{P})$ is given as

$$\mathcal{L}_{\delta}(\bar{\mathbf{P}}_{\sigma^*}, \mathbf{P}) = \mathcal{L}_{\delta,s}(\bar{\mathbf{P}}_{\sigma^*}, \mathbf{P}) + \mathcal{L}_{\delta,h}(\bar{\mathbf{P}}_{\sigma^*}, \mathbf{P}), \quad (13)$$

where $\mathcal{L}_{\delta,s}(\bar{\mathbf{P}}_{\sigma^*}, \mathbf{P}) = -\sum_{\ell=1}^L \log(\bar{s}_{\sigma(\ell)})$ is the negative log-likelihood loss that forces the score value of a valid path prediction goes to 1 and $\mathcal{L}_{\delta,h}(\bar{\mathbf{P}}_{\sigma^*}, \mathbf{P}) = \sum_{\ell=1}^L (|\bar{\beta}_{\sigma^*(\ell)} - \beta_{\ell}| + |\bar{\tau}_{\sigma^*(\ell)} - \tau_{\ell}| + |\bar{\phi}_{\sigma^*(\ell)} - \phi_{\ell}| + |\bar{\theta}_{\sigma^*(\ell)} - \theta_{\ell}|)$. The network parameters are updated as $\delta_j = \delta_{j-1} - \eta \nabla_{\delta} \mathcal{L}_{\delta,j-1}$ where η is the learning rate.

IV. SIMULATION AND DISCUSSIONS

A. Simulation Setup

We consider mmWave MISO systems where BS equipped with $N = 4 \times 4$ UPA antennas serves a single-antenna user with $f_c = 28$ GHz. To simulate wireless environment obtain the RGB images of the environment, we use Unity game engine. The human holding mobile device with an arbitrary motion is randomly located within the 8 m cell and moves straight with random speed. For large-scale fading coefficient $\beta = \rho(r) \cdot 10^{\frac{z\sigma_{\text{sh}}}{10}}$, we use path loss and shadow fading model specified in 3GPP TR 38.901 Rel. 17 [10], where $\rho(r) = -(31.84 + 21.5 \log_{10}(r) + 19 \log_{10}(f_c/10^9))$ dB, $\sigma_{\text{sh}} = 4$, and $z_k \sim \mathcal{N}(0, 1)$. We use MMSE channel estimation method for input and exploit maximum ratio transmission (MRT) method yielding $\mathbf{f}_t = \mathbf{h}_t$.

We compare SMCPT with three conventional methods: 1) Transformer-based channel prediction technique [11], 2) reciprocity-based channel prediction technique [12], and 2) instantaneous MMSE-based channel estimation.

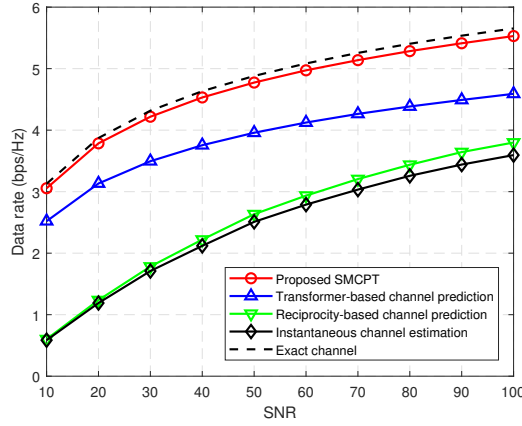


Fig. 5: Data rate vs. received SNR (user speed = 30 km/h).

B. Simulation Results

In Fig. 4, we plot the normalized MSE (NMSE) of the predicted channel as a function of the user speed when received SNR is 10. We can easily see that the proposed SMCPT achieves a significant accuracy improvement compared to the conventional channel prediction techniques by a large margin with every user speed. For instance, when user speed is 80 km/h, SMCPT achieves a reduction of more than 69%, 79%, and 83% in NMSE over the conventional Transformer-based channel prediction, reciprocity-based channel prediction, and instantaneous channel estimation, respectively. Also, we can observe that while the conventional schemes suffer from large channel prediction error with high user speed due to highly varying channels, the proposed SMCPT can limit the channel prediction error even with high user speed. This is because SMCPT can detect sudden changes in channels using the RGB cameras that quickly capture wireless environment.

In Fig. 5, we plot the data rate as a function of the received SNR. We observe that SMCPT outperforms the conventional channel prediction techniques by a large margin. For example, when $\text{SNR} = 50$, SMCPT achieves more than 20%, 81%, and 90% in data rate improvements over the conventional Transformer-based channel prediction, reciprocity-based channel prediction, and instantaneous channel estimation, respectively. This is because using the RGB images, SMCPT can precisely capture sudden changes in wireless environment. By exploiting an accurate channel information, we can properly align the beam with the channel, achieving 97% of the data rate obtained in the ideal case.

V. CONCLUSION

In this paper, we proposed the sensing-aided multi-modal channel prediction technique called SMCPT for mmWave systems. In contrast to the conventional approaches relying exclusively on the RF pilot-based channel estimations, we combine camera sensing with historical channel information to recognize the environment precisely. By extracting the positions and orientations of mobile devices, humans, and other entities affecting the signal propagation, SMCPT can

accurately predict the future channels. Furthermore, due to the short shooting duration of RGB cameras, SMCPT can immediately analyze wireless environment in a short time, thereby tracking the channel in a real-time manner.

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